

ACTU 301 ASSIGNMENT 1

Q1

Consider a policy which consists of a payment of \$50,000 contingent upon retirement in 15 years (if the person is still alive). Suppose that the probability of a 45-year old to retire in 15 years is 0.82. That is ${}_{15}p_{45} = 0.82$. Find the actuarial present value of the contingent payment. Assume, a 6% interest compounded annually.

Q2

You are currently 45years old. It is found that your mortality follows De Moivre's law with $\omega = 90$. You purchase a whole life insurance policy that pays a benefit of \$1,000,000 at the moment of death. Calculate the actuarial present value of your death benefits. Assume an annual effective interest rate of 10%.

Q3

The pdf of a future-lifetime random variable for a life aged x is given by

$$f_T(t) = \mu e^{-\mu t}, \quad t \geq 0.$$

Assume a constant force of interest δ , find $\bar{A}_x$, ${}^2\bar{A}_x$, $Var(\bar{Z}_x)$

Q4

You are given the following information:

- A unit benefit n -year term life insurance policy
- The age at death random variable with exponential parameter μ .
- The constant force of interest δ

Find the Actuarial present value and its variance.

Q5

Suppose that the age at death random variable is exponential with constant force of mortality μ . Let $\bar{Z}_{x:\overline{n}|}^1$ be the present value of an n-year pure endowment for a life aged (x) with benefit payment of 1. Assume force of interest δ , find $A_{x:\overline{n}|}^1$, ${}^2A_{x:\overline{n}|}^1$ and $Var(\bar{Z}_{x:\overline{n}|}^1)$

Q6

The lifetime of a group of people has the following survival function associated with it $s(x) = 1 - \frac{x}{100}$ $0 \leq x \leq 100$. Paul, a member of the group is currently 40 years old and has a 15-year endowment insurance policy, which will pay him \$50,000 upon death. Find the actuarial present value of this policy. Assume an annual force of interest $\delta = 0.05$.

Q7

Let the remaining lifetime at birth random variable X be uniform on $[0, \omega]$. Let ${}_m\bar{Z}_x$ be the contingent payment random variable for a life aged x. Find ${}_m\bar{A}_x$ if the force of interest is δ .